Algorithmic Trading

Robert Kissell

Case Study 02: Pre-Trade Analysis

Assignment:

Perform pre-trade analysis for a trade list of twenty-five (25) stocks.

Use the following MI Parameters for your analysis:

a1 = 883.58; a2 = 0.35; a3 = 0.75; a4 = 0.82; b1 = 0.96

Data File: CaseStudy_PreTrade.csv

Project Description:

You are provided with a list of twenty-five (25) stocks. The client asks you to perform a pre-trade analysis for each stock in the list so that they can evaluate and compare different trading strategies.

The trade list is in the file: CaseStudy_PreTrade.csv

- In industry, the client will provide you with their trade list (Symbol, Side, Shares).
- You (B/D) will populate the market data fields from a data source (such as Bloomberg, FactSet, S&P Capital IQ, etc.) or company database. These fields include Price, Volatility, ADV.
- The price trend estimate, AlphaBp, is provide by the client and is estimated via a proprietary alpha model.
    - The AlphaBp can also be calculated from the stock's trade beta and expected market movement on the day.

Calculate the following for each stock:

- Using a POV = 10%, Calculate MI, TR, PA, and Trade Time for each stock.
- Traders Dilemma = Calculate the Optimal POV Rate using Lambda = 0.25
- Minimize Cost = Calculate the Optimal POV Rate using the AlphaBp provided for each stock.
- Price Improvement = Calculate the Optimal POV Rate using Bid=100bp for each stock.
- Save the results in a csv file.

To Submit:

i) Upload a Table with your results. The table can be submitted as a CSV file or as an Excel spreadsheet.

ii) Upload your Python notebook